

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

Department: Computer Science and Engineering

Program: B.Sc. in Computer Science and Engineering

Semester Final Examination: Fall 2020

Year: 4th**Semester: 1st**

Course Number: CSE4107

Course Name: Artificial Intelligence

Time: 2 (Two) Hours

Full Marks: 50

Use single answer script.

Instructions	i)	Answer scripts should be handwritten and should be written in A4 white paper. You must submit the hard copy of the answer script to the Department when the university reopens.
	ii)	You must write the following information on the top page of your answer script: Department: Course no.: Examination: Student ID: Program: Course Name: Semester (Session): Signature and Date:
	iii)	Write down Student ID, Course number and put your signature on top of every single page of the answer script.
	iv)	Write down page number at the bottom of every page of the answer script.
	v)	Upload the scan copy of your answer script in PDF format through provided google form at the respective course site (i.e., google classroom) using institutional email within the allocated time. Uploading clear and readable scan copy (uncorrupted) is your responsibility and you must cover all the pages of your answer script. However, for clear and readable scan copy of the answer script student should use only one side of a page for answering the questions.
	vi)	You must avoid plagiarism and, maintain academic integrity and ethics . You are not allowed to take any help from another individual. If you do so, it can result in stern disciplinary actions against you from the university authority.
	vii)	Marks allotted are indicated in the right margin .
	viii)	Assume any reasonable data if needed.
	ix)	Symbols and characters have their usual meaning.
	x)	Before uploading, rename the PDF file as follows. CourseNo_StudentID Example: CSE4107_170204025.pdf

There are **6 (Six)** questions, answer **any 4 (Four)**.

Question 1. [Marks: 12.5]

- | | | |
|----|--|-------|
| a) | Derive a most general unifier for the following set of FOL sentences.
$S = \{P_i(x, F1(\text{Toy}), y, u), P_i(\text{Happy}, z, F2(x), v)\}.$ | [4] |
| b) | Analyze the role of issues related to environment of functioning in designing rational agents. | [4.5] |
| c) | Present a comparative study of the forward chaining and backward chaining algorithms. | [4] |

Question 2. [Marks: 12.5]

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|----------|--|-------|----|----|----|----|----|----|----|----|----|---|---|----------|---|---|---|---|---|---|---|---|---|---|---|--------|----|----|----|----|----|----|----|----|----|----|----|------|---|---|---|---|---|---|---|---------|----|----|----|----|----|----|---|-----|
| a) | <p>Given below through the tables is a weighted undirected graph along with the heuristic function values of the nodes.</p> <table border="1"> <tr><td>Node</td><td>A</td><td>A</td><td>B</td><td>B</td><td>B</td><td>C</td><td>C</td><td>D</td><td>D</td><td>E</td><td>F</td></tr> <tr><td>Neighbor</td><td>B</td><td>C</td><td>C</td><td>D</td><td>E</td><td>D</td><td>F</td><td>E</td><td>G</td><td>G</td><td>G</td></tr> <tr><td>Weight</td><td>45</td><td>42</td><td>37</td><td>36</td><td>28</td><td>26</td><td>32</td><td>24</td><td>47</td><td>36</td><td>41</td></tr> </table>
<table border="1"> <tr><td>Node</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td><td>G</td></tr> <tr><td>h(Node)</td><td>70</td><td>50</td><td>52</td><td>38</td><td>33</td><td>30</td><td>0</td></tr> </table> <p>Show that the heuristic function is consistent.</p> | Node | A | A | B | B | B | C | C | D | D | E | F | Neighbor | B | C | C | D | E | D | F | E | G | G | G | Weight | 45 | 42 | 37 | 36 | 28 | 26 | 32 | 24 | 47 | 36 | 41 | Node | A | B | C | D | E | F | G | h(Node) | 70 | 50 | 52 | 38 | 33 | 30 | 0 | [4] |
| Node | A | A | B | B | B | C | C | D | D | E | F | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Neighbor | B | C | C | D | E | D | F | E | G | G | G | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Weight | 45 | 42 | 37 | 36 | 28 | 26 | 32 | 24 | 47 | 36 | 41 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Node | A | B | C | D | E | F | G | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| h(Node) | 70 | 50 | 52 | 38 | 33 | 30 | 0 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| b) | Present an analysis of the biological inspiration deployed in common Genetic algorithms. | [4.5] | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| c) | Demonstrate a comparison between A* search and Hill-climbing local search. | [4] | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

Question 3. [Marks: 12.5]

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|----|--|-------|
| a) | Construct a pruned game tree using Alpha-Beta pruning. Take the sequence, [5, 3, 2, 4, 1, 3, 6, 2, 7, 8, 5, 4, 1, 6] of MINIMAX values for the nodes at the cutoff depth of 4 plies and assume that branching factor is 2, MIN makes the first move, and nodes are generated from <i>right to left</i> . | [4] |
| b) | Analyze the characteristics of a planner to establish that it is an intelligent agent. | [4.5] |
| c) | Compare the features of search strategies used for planners and for game playing algorithms. | [4] |

Question 4. [Marks: 12.5]

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|----------|---|------|---|------|---|---|-----|---|----------|-----|----------|---|-----|----------|----------|------|---|---|------|---|---|-----|---|----------|-----|----------|---|-----|----------|----------|------|---|------|---|-----|----------|------|------|------|-----|
| a) | <p>Say, we have a Bayesian network containing 4 random variables A, B, C and D with two distinct values each. The conditional probability tables assigned to the random variables are given below. Draw the network and compute $P(C \mid a \wedge \neg b \wedge \neg d)$ using the network.</p> <table border="1"> <tr><td>A</td><td>B</td><td>P(d)</td></tr> <tr><td>a</td><td>b</td><td>0.9</td></tr> <tr><td>a</td><td>$\neg b$</td><td>0.6</td></tr> <tr><td>$\neg a$</td><td>b</td><td>0.3</td></tr> <tr><td>$\neg a$</td><td>$\neg b$</td><td>0.01</td></tr> </table> <table border="1"> <tr><td>A</td><td>D</td><td>P(c)</td></tr> <tr><td>a</td><td>d</td><td>0.8</td></tr> <tr><td>a</td><td>$\neg d$</td><td>0.4</td></tr> <tr><td>$\neg a$</td><td>d</td><td>0.2</td></tr> <tr><td>$\neg a$</td><td>$\neg d$</td><td>0.02</td></tr> </table> <table border="1"> <tr><td>B</td><td>P(a)</td></tr> <tr><td>b</td><td>0.7</td></tr> <tr><td>$\neg b$</td><td>0.04</td></tr> </table> <table border="1"> <tr><td>P(b)</td></tr> <tr><td>0.08</td></tr> </table> | A | B | P(d) | a | b | 0.9 | a | $\neg b$ | 0.6 | $\neg a$ | b | 0.3 | $\neg a$ | $\neg b$ | 0.01 | A | D | P(c) | a | d | 0.8 | a | $\neg d$ | 0.4 | $\neg a$ | d | 0.2 | $\neg a$ | $\neg d$ | 0.02 | B | P(a) | b | 0.7 | $\neg b$ | 0.04 | P(b) | 0.08 | [4] |
| A | B | P(d) | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| a | b | 0.9 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| a | $\neg b$ | 0.6 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| $\neg a$ | b | 0.3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| $\neg a$ | $\neg b$ | 0.01 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| A | D | P(c) | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| a | d | 0.8 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| a | $\neg d$ | 0.4 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| $\neg a$ | d | 0.2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| $\neg a$ | $\neg d$ | 0.02 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| B | P(a) | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| b | 0.7 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| $\neg b$ | 0.04 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| P(b) | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 0.08 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

b)	Analyze the evaluation steps of a Decision network to justify that the basic principle of decision theory underlies the whole process.	[4.5]
c)	Briefly demonstrate a comparison between inference using full joint probability distribution and inference using Bayes' theorem.	[4]

Question 5. [Marks: 12.5]

a)	Consider the 'Escape the monster' environment given below. <table border="1" data-bbox="248 544 603 777"> <tr> <td>OK 1,2</td><td>OK 2,2 S</td><td></td></tr> <tr> <td>OK 1,1</td><td>OK 2,1</td><td>OK 3,1 S</td></tr> </table> Assume that OK indicates absence of a monster(M) at the cell, the status of the cells of the 4 x 4 grid other than those shown are unknown, and that the agent can smell(S) a monster at a cell vertically or horizontally adjacent to it. Find $P(M_{4,1})$ taking 0.3 as the independent probability of a monster at any of the unknown cells.	OK 1,2	OK 2,2 S		OK 1,1	OK 2,1	OK 3,1 S	[4]
OK 1,2	OK 2,2 S							
OK 1,1	OK 2,1	OK 3,1 S						
b)	How do you interpret the final output of a typical Decision Tree learning algorithm?	[4.5]						
c)	Evaluate the advantages and disadvantages of Neural Network learning with respect to other machine learning algorithms.	[4]						

Question 6. [Marks: 12.5]

a)	Given below is a table that describes some 3-attribute training samples with their class tags. <table><tr><th>A1</th><th>A2</th><th>A3</th><th>Class</th></tr><tr><td>1</td><td>H</td><td>a</td><td>N</td></tr><tr><td>1</td><td>L</td><td>a</td><td>P</td></tr><tr><td>1</td><td>H</td><td>b</td><td>N</td></tr><tr><td>2</td><td>H</td><td>a</td><td>P</td></tr><tr><td>2</td><td>L</td><td>a</td><td>N</td></tr><tr><td>2</td><td>H</td><td>b</td><td>N</td></tr></table> Find a Naïve Bayes prediction for the sample, (1, L, b).	A1	A2	A3	Class	1	H	a	N	1	L	a	P	1	H	b	N	2	H	a	P	2	L	a	N	2	H	b	N	[4]
A1	A2	A3	Class																											
1	H	a	N																											
1	L	a	P																											
1	H	b	N																											
2	H	a	P																											
2	L	a	N																											
2	H	b	N																											
b)	How do you characterize the application domains of clustering algorithms?	[4.5]																												
c)	Present a brief comparison between the use of k-Nearest Neighbor classifiers and the use of Support Vector machines as classifiers.	[4]																												